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Nitrogen fixation in endophytic and associative symbiosis

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Abstract

N-balance, ^{15}N isotope dilution and ^{15}N natural abundance studies provide strong evidence that some tropical grasses, especially sugar cane (*Saccharum* spp.), wetland rice (*Oryza sativa*) and kallar grass (*Leptochloa fusca*) can obtain at least part of their N-needs from biological nitrogen (N_2) fixation. However, these studies have not provided conclusive evidence that these plants are engaged in symbiotic partnerships with any bacteria, as the techniques used do not distinguish between “symbiotic” N_2 fixation on the one hand, and N_2 fixation inputs from free-living heterotrophs and cyanobacteria on the other. Large and diverse populations of heterotrophic diazotrophs can be isolated from the surfaces and rhizospheres of sugar cane, wetland rice and kallar grass, and also from surface-sterilized plants. While the latter suggests that these organisms are living intimately within the plant tissues and hence may be fixing N_2 in symbiotic partnerships with their hosts, “endophytic diazotrophs” have been observed only within the intercellular spaces, vascular tissue, aerenchyma and dead cells of their hosts, not in their living cells. Recent studies have also shown expression by diazotrophs of genes encoding for nitrogenase (e.g., *nifA*, *nifH*), in situ expression of the nitrogenase enzyme proteins, and nitrogenase (acetylene reduction) activity in association with their host plants, but have yet to determine whether the bacteria transfer the products of N_2 fixation to the plants. Earlier studies used $^{15}\text{N}_2$ gas incorporation to show transfer of fixed N in sugar cane, rice, maize (*Zea mays*), sorghum (*Sorghum bicolor*) and various forage grasses. However, the long exposure times often used in these studies could mean that the fixed N may have been transferred principally by the death and mineralization of asymbiotic diazotrophs, rather than through direct and rapid transfer as occurs in legume nodules. To demonstrate that N_2 -fixing symbioses are present, future $^{15}\text{N}_2$ experiments should utilize short exposure times (hours), and compare plants inoculated with wild type diazotrophs and *nif* mutants. Only an integrated and multidisciplinary approach can determine if (a) N_2 -fixing symbioses with Gramineae really exist, and (b) if they are effective in agronomic terms. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Nitrogen fixation; Gramineae; Sugar cane; Rice; Kallar grass; Endophytic diazotrophs; *Acetobacter diazotrophicus*; *Azoarcus*; *Azospirillum*; *Herbaspirillum*

1. Introduction

Recent evidence of significant biological nitrogen (N_2) fixation in economically important Gramineous species, particularly sugar cane (*Saccharum* sp.)

(Urquiaga et al., 1992), rice (*Oryza sativa*) (Shrestha and Ladha, 1996) and forage grasses, such as kallar grass (*Leptochloa fusca*) (Malik et al., 1997) has (re)generated tremendous interest in N_2 fixation by non-legumes (see reviews by Boddey, 1995; Boddey et al., 1995, 1999; Ladha and Reddy, 1995; Triplett, 1996; Kennedy et al., 1997; Reinhold-Hurek and

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Hurek, 1998a,b). Unlike most studies in the 1960s, 1970s and 1980s that largely concentrated their attention on rhizosphere bacteria as likely sources of fixed N, much of the recent interest has focussed upon the possibility that bacteria within the plants themselves, i.e., “endophytic diazotrophs” (Döbereiner et al., 1995), are responsible for the observed N_2 fixation. The most likely candidates for BNF in grasses are *Acetobacter diazotrophicus* and *Herbaspirillum* spp. in sugar cane (Döbereiner et al., 1995; Boddey et al., 1995, 1999; Baldani et al., 1997), *Azoarcus* spp. in kallar grass (Reinhold-Hurek and Hurek, 1998a,b), and species of *Alcaligenes*, *Azospirillum*, *Bacillus*, *Enterobacter*, *Herbaspirillum*, *Klebsiella*, *Pseudomonas* and *Rhizobium*, in rice and maize (*Zea mays*) (Patriquin et al., 1983; Boddey et al., 1995; Triplett, 1996; Malik et al., 1997; Stoltzfus et al., 1997; Yanni et al., 1997; James et al., 1999).

James and Olivares (1998) briefly reviewed the evidence that endophytic diazotrophs may actually fix N_2 in planta and transfer the fixed N products to their hosts. The latter point is an essential feature of any N_2 -fixing symbiosis. For example, it has been clearly demonstrated using a range of techniques that the symbioses between legumes/rhizobia, actinorhizal plants/*Frankia*, *Azolla*/*Anabaena* and *Gunnera*/*Nostoc* all involve a significant net transfer of N from the microsymbionts to the host in return for photosynthetically derived carbon substrates (see reviews by Sprent and Sprent, 1990; Mylona et al., 1995). James and Olivares (1998) concluded that there was, as yet, no real evidence to support the suggestion that associations between N_2 -fixing rhizosphere bacteria (endo or epiphytic) and Gramineae are also symbiotic. Indeed, if associative/endophytic “symbioses” exist they must be quite different from most previously described plant/diazotroph symbioses where symbiotic “organs” have evolved to house large concentrations of heterotrophic bacteria, or as with *Azolla*, specialized leaf cavities (Sprent and Sprent, 1990; Mylona et al., 1995).

The purpose of this paper is to update and expand upon some of the issues raised by James and Olivares (1998) with regard to evidence for N_2 fixation by non-legumes, particularly in the Gramineae. It focuses primarily on recent evidence that the N_2 fixation in the grasses may be symbiotic, and examines the methods that have been, or could be, utilized to prove this.

2. Field and greenhouse evidence

Arguably, field studies give the most important evidence for N_2 fixation. Although greenhouse and laboratory studies can provide useful academic/scientific information, significant yield increases in field crops due to N_2 fixation are the ultimate aim of all researchers in this area. Growth increases/N-accumulation associated with apparent nitrogenase activity as shown by the acetylene reduction assay have been the mainstays of research in this area and there have been a vast number of studies in which crop response to inoculation with *Azospirillum* has been evaluated (Sumner, 1990; Okon and Labandera-Gonzalez, 1994). They have been inconclusive as it is impossible to say with certainty that the observed effects (if any) of such inoculation are due to plant-associated N_2 fixation, and not to factors, such as hormonal effects, enhanced scavenging of N from the soil due to bacterial-induced increases in root growth, and/or bacterial nitrate reductase (Michiels et al., 1989; Sumner, 1990; Bashan and Levanony, 1990).

Drawbacks in the acetylene reduction assay have been discussed by van Berkum and Bohlool (1980) and Boddey (1987). Although this technique can be a useful “spot-check” of nitrogenase activity at the time the assay is conducted, difficulties arise when extrapolations about actual amounts of N fixed are made from the data, as the assay only measures nitrogenase activity and says nothing about whether the fixed N is incorporated into the plant (Boddey, 1987; Boddey et al., 1995). It is because of this that most workers now agree that little useful information about N_2 fixation in non-legumes can be obtained if acetylene reduction is the sole means of evaluation used.

2.1. ^{15}N isotope dilution and ^{15}N natural abundance studies

Careful N-balance experiments can provide strong indications that a crop is obtaining some of its N content from N_2 fixation, but are very laborious and time-consuming. Their interpretation may also suffer from the fact that plant N, particularly in the case of wetland rice, could have been derived from a number of sources, including (non-associative/asymbiotic) N_2 fixation by cyanobacteria in the soil (Boddey, 1987; Boddey et al., 1995; Ladha and Reddy, 1995; Gantar

et al., 1995; Kamuru et al., 1998), or from atmospheric ammonia, irrigation water and rainfall (Boddey, 1987). Accordingly, the most useful methods for examining N_2 fixation in the field and in large greenhouse experiments are still the ^{15}N isotope dilution and ^{15}N natural abundance techniques. In the former, plants are grown in soil enriched in ^{15}N -containing compounds (usually ammonia, nitrate or urea). Since microbially fixed N dilutes ^{15}N taken up from soil, those plants that have a significant N input from N_2 fixation should have a lower concentration of ^{15}N in their tissues compared to “non-fixing” control plants. The differential between N_2 -fixing and control plants can be measured using mass spectrometry. The ^{15}N natural abundance technique utilizes the fact that soil is naturally slightly enriched in ^{15}N compared to the atmosphere, and plants in which a significant amount of N is derived from fixation should have a measurably lower concentration of ^{15}N in their tissue. Boddey (1987), Shearer and Kohl (1986), Chalk (1991) and Högburg (1997) describe practical applications of both these techniques and the problems involved in their use.

Both ^{15}N isotope dilution and ^{15}N natural abundance studies have shown that sugar cane can fix substantial amounts of N_2 (Urquiaga et al., 1992; Boddey et al., 1995, 1999; Yoneyama et al., 1997), but also that the amount fixed is highly variable and is dependent on plant genotype and environmental conditions. Indeed, significant N_2 fixation has only been shown by the Brazilian sugar cane varieties CB 45-3, SP 70-1143, and the *S. spontaneum* variety Krakatau (Urquiaga et al., 1992). Although these studies were actually performed in a large concrete tank filled with ^{15}N -labelled soil, the quasi-field nature of this work has made it possible to reasonably estimate that such varieties can fix up to 70% of their N-requirements, and provide over 150 kg N fixed ha^{-1} per year (Boddey et al., 1995). It has been suggested that the relatively low mineral N inputs used for Brazilian cane production over the last 100 years has inadvertently selected for varieties with a low response to applied mineral N and a high N_2 -fixing ability (Ruschel, 1981; Boddey, 1995). However, it must also be stressed that N_2 fixation by Brazilian sugar cane seems to depend on optimizing conditions of water supply, P and K availability, and micronutrients, especially molybdenum (Urquiaga et al., 1992, 1995).

Although most of their data was obtained in Brazil, Yoneyama et al. (1997) have also reported N_2 fixation from sites in Japan and the Philippines using the ^{15}N natural abundance technique. Clearly, ^{15}N studies are needed to determine if N_2 fixation can occur in other sugar cane-producing countries, particularly those which have high levels of mineral N-inputs, such as Mexico, India, Cuba, Australia and the USA (Ruschel, 1981; Boddey, 1995; Fuentes-Ramirez et al., 1999).

Compared to sugar cane, recent field- and greenhouse studies with other grasses have given generally less promising, and rather more variable results. For example, using similar methodology to Urquiaga et al. (1992), i.e., a large concrete tank, but this time filled with waterlogged soil, Boddey et al. (1995) failed to confirm N_2 fixation contributions to three wetland rice varieties. On the other hand, recent ^{15}N isotope dilution greenhouse studies (Wu et al., 1995; Shrestha and Ladha, 1996) at the International Rice Research Institute (IRRI) have shown that some rice genotypes can obtain significant quantities of N from N_2 fixation. Seventy varieties examined by Shrestha and Ladha (1996) gave nitrogen derived from air (Nd_fa) values ranging from 0 to 20.2%. This is in agreement with many ^{15}N isotope dilution and N-balance studies previously reported with rice (see reviews by Chalk, 1991; Boddey et al., 1995, 1999; Ladha and Reddy, 1995), and suggests that BNF can make a significant, albeit highly variable, contribution to the N-budget of areas with low mineral N inputs. Although a number of ^{15}N isotope dilution studies by Rennie and co-workers in the 1980s showed quite significant N_2 fixation by wheat (*Triticum aestivum*) (e.g., Rennie et al., 1983), these claims have since been disputed (see reviews by Boddey, 1987, and Chalk, 1991). Indeed, more recent studies have shown little or no N_2 fixation by this plant, with or without inoculation by *Azospirillum* and other diazotrophs (Kapulnik et al., 1985; Boddey, 1987; Chalk, 1991; Bremer et al., 1995). Although *Azospirillum* and *Azotobacter beijerinckia* contributed minor amounts of fixed N (Lethbridge and Davidson, 1983; Christiansen-Weniger et al., 1992), Lethbridge and Davidson (1983) concluded that the bacteria probably only transferred fixed N to the plant when they died and were subsequently mineralized. Indeed, much of the “plant-associated” N_2 fixation reported from ^{15}N isotope

dilution studies could be due to this process, a point discussed further in Section 5.

Compared to sugar cane, rice and wheat, there have been relatively few studies on N_2 fixation with maize and sorghum. Recent work again suggests that N_2 fixation in these crops may be significant, but also that genotypic variation is normal. For example, in a study of 15 maize varieties, Garcia de Salamone and Döbereiner (1996) showed that some varieties had significant increases in grain yield and N-content after inoculation with mixed strains of *Azospirillum* spp., whereas other varieties showed decreases. In a subsequent ^{15}N isotope dilution study using pots maintained in the field, Garcia de Salamone et al. (1996), showed that two of the maize varieties in their previous study (Dekalb D4-70 and CMS22) could fix up to 58% of their N if inoculated with appropriate strains of *Azospirillum* spp. On the other hand, in the case of sorghum, using ^{15}N natural abundance, Sarig et al. (1990) and Lee et al. (1994) observed no significant BNF in either the controls or the *Azospirillum*/*Azotobacter*-inoculated plants. The large margins of error inherent in this methodology (Shearer and Kohl, 1986; Boddey, 1987) could have obscured a small contribution by N_2 fixation. Other studies, using ^{15}N isotope dilution and $^{15}N_2$ gas incorporation, have shown some (although relatively minor) potential for N_2 fixation in both *Azospirillum*-inoculated, and uninoculated sorghum (Giller et al., 1984; Boddey, 1987).

In Brazil, a number of tropical forage grasses, including *Brachiaria humidicola*, *B. decumbens*, *Paspalum notatum* and *Panicum maximum* have shown relatively high N_2 fixation rates in ^{15}N isotope dilution studies, and may derive up to 40% of their N-needs from fixation (Boddey, 1987; Miranda et al., 1990). Nitrogen fixation by grasses in the USA and Canada (Boddey, 1987), and by kallar grass in Pakistan have also been reported (Malik et al., 1997, and see Section 6).

3. Isolation of the organisms responsible for N_2 fixation in grasses

Although ^{15}N isotope dilution/natural abundance studies have given much useful information on the potential for N_2 fixation of non-legumes, they have not generally provided information on the causal organisms. The simple isolation of diazotrophs (e.g., *A.*

diazotrophicus, *Herbaspirillum* spp.) from a plant (e.g., sugar cane) is, by itself, inadequate evidence for plant-associated BNF. A great variety of bacteria, diazotrophs and non-diazotrophs, can be isolated from the rhizosphere and endorhizosphere of sugar cane using appropriate N-free media (Rennie et al., 1982; Li and MacRae, 1992; Boddey et al., 1995; Baldani et al., 1997; Fuentes-Ramirez et al., 1999). Although *A. diazotrophicus* and *Herbaspirillum* spp. appear to be the most numerous of the bacteria so far isolated from N_2 -fixing varieties of sugar cane (Döbereiner et al., 1995; Boddey et al., 1995, 1999; Baldani et al., 1997), as correlations between N_2 -fixing ability and bacterial numbers have yet to be presented, it is currently impossible to attribute N_2 fixation in this plant to any specific bacterium (James and Olivares, 1998; Boddey et al., 1999). Nevertheless, Dong et al. (1994) have claimed a “symbiosis” between sugar cane and *A. diazotrophicus*, even though they isolated *A. diazotrophicus* from sugar cane varieties (cvs. Ja 60, Media Luna) not previously evaluated for active N_2 fixation, and presented no evidence of plant-associated N_2 fixation. As detailed above, the studies of Lima et al. (1987), Urquiaga et al. (1992), and Yoneyama et al. (1997), showed evidence of BNF in only a few, mainly Brazilian, sugar cane varieties. It is simplistic to assume that a high proportion of sugar cane varieties collected from all over the world fix N_2 . Field work to identify a range of varieties with the ability to fix N_2 must be undertaken, and the agronomic and environmental factors likely to affect fixation should be evaluated. The systematic isolation and enumeration of the endophytic and associative diazotroph populations in varieties showing different N_2 fixation abilities should be performed in parallel with the field work.

Many of the above arguments also apply to rice and other grasses. However, in the case of wetland rice, even when specific varieties have been shown to fix N_2 (Wu et al., 1995; Shrestha and Ladha, 1996), it will be extremely difficult to isolate the organisms responsible, because

1. Approximately 90% of the bacteria isolated from surface-sterilized rice plants (several species and varieties, plus some related genera) using N-deficient media are non-diazotrophs (Barraquio et al., 1997).

2. The culturable diazotrophic population is extremely varied, and so far virtually uncharacterized (Barraquio et al., 1997; Stoltzfus et al., 1997; Yanni et al., 1997).
3. The rhizosphere of rice may also contain an enormous bacterial population that has yet to be cultured (Ueda et al., 1995a,b; Reinhold-Hurek and Hurek, 1998a,b).

4. Localizing the diazotrophs responsible for N₂ fixation in grasses

If diazotrophs can be isolated from the rhizosphere and/or the interior of plants in which N₂ fixation is significant, and the bacteria are present in or on the host, it should be possible to observe them in situ. Indeed, considerable progress has been made in recent years in localizing potentially important diazotrophs on and within non-legumes using both optical and electron microscopy (James and Olivares, 1998; Reinhold-Hurek and Hurek, 1998a,b). Localization of diazotrophs has been greatly facilitated through the use of immunogold and immunofluorescence labelling, utilizing either polyclonal (e.g., Levanony et al., 1989; James et al., 1994, Fig. 1a–c) or monoclonal (e.g., Assmus et al., 1997; Schlöter and Hartmann, 1998) antibodies. Moreover, constitutively expressed bacteria-reporter gene-fusions, such as β -galactosidase (*lacZ*) (Katupitiya et al., 1995; Gough et al., 1997; Reddy et al., 1997), β -glucuronidase (*GUS*) (Hurek et al., 1994; Gough et al., 1997; Fuentes-Ramirez et al., 1999), green fluorescent protein (*gfp*) (Egener et al., 1998), as well as fluorescently labelled, species-specific rRNA-targeted oligonucleotide probes (Assmus et al., 1995, 1997), are powerful tools that are being increasingly used in localization studies. Indeed, it is now considered that labelling is essential in microscopical studies of associative plant–bacterial interactions (particularly those grown in field or other non-sterile situations). Plants can contain many different types of bacteria, impossible to distinguish using conventional microscopy (Levanony et al., 1989; Hallmann et al., 1997; James and Olivares, 1998; Reinhold-Hurek and Hurek, 1998a,b). In addition, Reinhold-Hurek and Hurek (1998a,b) have further suggested that only studies performed on either samples that are intact, such as those used in scanning

confocal laser microscopy (SCLM; e.g., Assmus et al., 1995; Reddy et al., 1997), or on samples that have been chemically fixed and embedded in a permanent and stable medium, such as a resin or a plastic, can be considered reliable evidence for the exact location of the diazotrophs being studied.

Most non-endophytic, associative, diazotrophs, e.g., some strains of *Azospirillum*, live either on the surface of the roots, particularly in the root hair and elongation zones, or within disrupted epidermises/outer cortices (see reviews by Patriquin et al., 1983; Michiels et al., 1989; Bashan and Levanony, 1990). In contrast, endophytic diazotrophs, such as *A. diazotrophicus*, *Azoarcus* spp., *Herbaspirillum* spp., and some strains of *Azospirillum brasilense* (e.g., strain Sp245; Schlöter and Hartmann, 1998), tend to colonize the root cortex, and may even penetrate the endodermis to colonize the stele, from which they may be subsequently translocated to the aerial parts. Endophytic diazotrophs do not live within healthy host cells but instead colonize the apoplast, i.e., the intercellular spaces (Fig. 1c), the xylem vessels (Fig. 1a and b) and lignified xylem parenchyma, as well as dead cells, such as those comprising lysigenous aerenchyma in rice (Fig. 1d) and kallar grass. A probable site of entry for endophytic bacteria are cracks formed at lateral root junctions, e.g., *H. seropedicae* infecting rice roots (Fig. 1c) (and see James et al., 1999). This and other infection mechanisms are discussed in more detail by Sprent and James (1995), Hallmann et al. (1997), James and Olivares (1998) and Reinhold-Hurek and Hurek (1998a,b).

4.1. Localization of *nif* genes and nitrogenase proteins

In addition to confirming the identity of the bacteria within samples, immunolabelling, in situ hybridization, and reporter genes can also be used to detect the expression of nitrogenase proteins and/or nitrogenase-encoding (*nif*) genes. An antibody raised against the iron protein of nitrogenase isolated from *Rhodospirillum rubrum* was used to detect *in planta* expression of nitrogenase in *Azoarcus* sp. BH72 associated with rice and kallar grass (Hurek et al., 1994; Reinhold-Hurek and Hurek, 1998a), and in *H. rubrisubalbicans* with sugar cane and sorghum (Olivares et al., 1997; James et al., 1997). A problem with this technique is

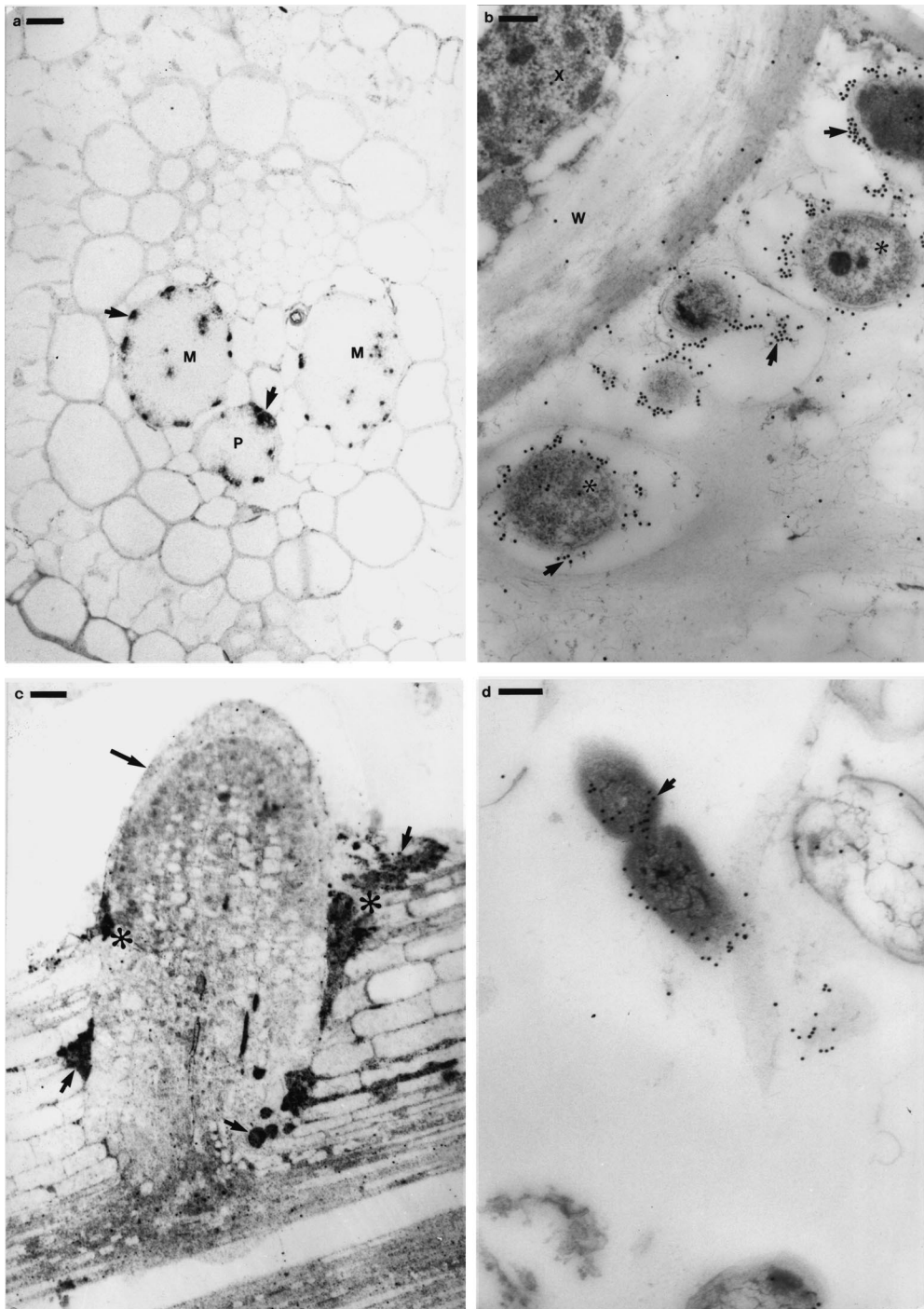


Fig. 1. (a) Optical micrograph of a transverse section through a sugar cane leaf (cv. SP 70-1143) 7 days after stem inoculation with *A. diazotrophicus* strain PAL-5. The section was immunogold-labelled with a polyclonal antibody raised against *A. diazotrophicus*, followed by silver-enhancement and light counterstaining with toluidine blue. The bacteria can be seen as black bodies (arrows) within the metaxylem (M) and the protoxylem (P). The bar=10 µm. (b) Transmission electron micrograph (TEM) of *A. diazotrophicus* (*) colonizing the protoxylem of a

that expression of nitrogenase proteins by bacteria *in planta* appears to be low compared to the free-living bacteria (Reinhold-Hurek and Hurek, 1998a) with the immunogold signal not always distinguishable from the background labelling. Nevertheless, as these studies show (see Fig. 1d), some useful data can still be obtained if steps are taken to reduce the background signal.

Reporter genes linked to various *nif* genes have been used to show expression of *nifA* (Arsene et al., 1994; Katupitiya et al., 1995) and *nifH* (Vande Broek et al., 1993) by *A. brasilense* on the surface of wheat roots. In the case of rice, Vermeiren et al. (1998) have shown the endophytic expression of *nifH* by *Alcaligenes faecalis* A15 and *Azospirillum irakense* within root epidermal cells, and Reinhold-Hurek and Hurek (1998b) and Egener et al. (1998) have obtained similar results with *Azoarcus*. *In planta* *Azoarcus nifH* expression was also demonstrated by in situ hybridization studies using antisense *nifH* probes applied to the roots of uninoculated kallar grass (Hurek et al., 1997). Recent work at IRRI has shown that a diazotrophic strain of *Serratia marcescens* isolated from wetland rice will express *nifH*, as well as the nitrogenase iron protein, whilst colonizing the aerenchyma and xylem vessels of inoculated rice seedlings (Fig. 1d, and see James et al., 1999). Interestingly, and in contrast to work with *Herbaspirillum* (Olivares et al., 1997), there is as yet no evidence of *nif* gene or nitrogenase protein expression by *A. diazotrophicus* in association with sugar cane, except for some acetylene reduction activity with inoculated plantlets (James et al., 1994). Indeed, a recent study by Fuentes-Ramirez et al. (1999) failed to detect *in planta* expression after

inoculation of sprouting setts by a *nifH-GUS* strain of *A. diazotrophicus*.

5. Transfer of the products of N₂ fixation to the host plant

Localization of diazotrophs in itself does not say much about a symbiosis. Even if relatively high concentrations of bacteria are observed associated, or within the plants, it does not mean that they are actually expressing nitrogenase. Furthermore, even if expression of *nif* genes and/or nitrogenase proteins is demonstrated, and acetylene reduction activity shown, it still does not establish that the products of N₂ fixation are being transferred to the host. The death and subsequent mineralization of diazotrophs can release significant quantities of fixed N (Lethbridge and Davidson, 1983; Lee et al., 1994) (see Section 5.1.), but this process is inefficient, and perhaps delayed, as compared to the active release of the immediate products of N₂ fixation by living bacteria (as occurs in legume nodules; Mylona et al., 1995). Plants which derive N products from the mineralization of associated diazotrophs (e.g., *Azospirillum*) cannot be correctly classed as “symbiotic” when they differ from “non-associative” plants deriving fixed N from mineralized free-living heterotrophs and/or cyanobacteria only in that the bacteria live on the roots, or in the rhizosphere immediately surrounding them. Moreover, none of the endophytic diazotrophs associated with sugar cane or other grasses have been found within the living host cells. Therefore, it may be that they are simply an extension of the soil/rhizo-

Fig. 1. (Continued)

sugar cane leaf. The section was immunogold-labelled with a polyclonal antibody raised against *A. diazotrophicus*. Note that there is intense labelling of the bacteria with 15 nm gold particles, particularly of the surface material (arrows). This material probably contains exopolysaccharide (EPS). X=xylem parenchyma; W=cell wall. The bar=200 nm. (c) Optical micrograph of a longitudinal section through a rice root 2 days after inoculation of the rooting medium with *Herbaspirillum seropedicae* strain Z67 transformed to constitutively express GUS (Barraquio et al., 1997). The section was immunogold-labelled with a polyclonal antibody raised against *H. seropedicae*, followed by silver-enhancement and light counterstaining with toluidine blue. The bacterial colonies can be seen as black bodies (arrows) adjacent to an emerging lateral root (large arrow). Note that the bacteria have extensively colonized intercellular spaces within the main root, and that the bacteria appear to be entering it via cracks at the junction formed by the emerging lateral root (*). The bar=20 µm. (d) TEM of a *Serratia* sp. colonizing the aerenchyma of a rice stem. This bacterium was originally isolated from surface-sterilized wetland rice plants, and subsequently transformed to contain a *nifH-GUS* fusion. The section was taken from an axenically grown plant at 5 days after inoculation with the *Serratia* sp. containing the *nifH-GUS* fusion. It was immunogold-labelled with an antibody raised against the iron protein of nitrogenase, and 15 nm gold particles can be seen associated with the bacteria (arrows). The bar=200 nm.

sphere bacterial population that has evolved to take advantage of the protective niches that the interior of the plants can provide. They cannot also be assumed to be providing fixed N₂ to the plants other than when they die and release fixed N either as amino compounds or ammonium.

Ammonium-excreting mutants of diazotrophs, particularly *Azospirillum*, have been produced, and these have been shown to benefit *para*-nodulated wheat and rice plants, presumably by fixing N₂ and rapidly transferring the fixed N products to the plants (Christiansen-Weniger, 1992; Kennedy et al., 1997). However, so far there is little evidence that wild types of common endophytic and/or root-associated diazotrophs have this ability, and it is unlikely that ammonium-excreting mutants would be competitive if released into the field. On the other hand, Cojho et al. (1993) have demonstrated that wild-type *A. diazotrophicus*, when co-cultured with an amyolytic yeast, can release up to 48% of its fixed N and make it available to the yeast. This suggests that *A. diazotrophicus* may be inherently “leaky” and in association may directly benefit its host.

5.1. ¹⁵N₂ gas incorporation

The transfer of fixed N from *A. diazotrophicus* to sugar cane could be demonstrated using ¹⁵N₂ gas incorporation (Boddey, 1987). In this technique a whole plant, or just its root system, is enclosed in an atmosphere amended with a small volume of ¹⁵N-labelled N₂. After suitable incubation, the plant material and/or the rhizosphere soil is examined for significant enrichment with ¹⁵N. Because of the relatively short time periods used, theoretically such enrichment can only be obtained through the transfer of fixed ¹⁵N to the plant and/or rhizosphere. Controls not exposed to ¹⁵N₂ are essential to determine if the incorporation is significant (Boddey, 1987). Drawbacks to the technique include the cost of the ¹⁵N₂ for assessing large plants such as sugar cane; this means that most studies have used limited replication. Other problems result from leakage in the system, which may result in an underestimate of the amount of N fixed and transferred (Boddey, 1987), and impurities in the ¹⁵N₂, particularly nitrogen oxides and ammonia, which can be assimilated by the plants and hence give overestimates of N₂ fixation (Bremer et al., 1995). Some of the

criticisms levelled at acetylene reduction methodologies can also be made of the ¹⁵N₂ technique. Firstly, it is a “spot” measurement used on only a few replicates at any one time and hence does not provide estimates of N₂ fixation over a growing season (Boddey, 1987; Boddey et al., 1995, 1999). Secondly, long incubation times, especially with the whole plant enclosed, can result in impaired photosynthesis and unusual growth (Giller et al., 1984; Boddey, 1987). Thirdly, delays between collection of samples from the field and their evaluation or long incubation times (often in lowered pO₂; Ruschel et al., 1975, 1981; de Polli et al., 1977) can also result in the substantial growth of free-living diazotrophs in the rhizosphere (van Berkum and Bohloul, 1980).

Nevertheless, even if the reservations of van Berkum and Bohloul (1980) are taken into consideration, a lot of interesting data has been obtained using the technique. It has provided evidence of N₂ fixation in sugar cane (Ruschel et al., 1975, 1981), rice (Ito et al., 1980; Yoshida and Yoneyama, 1980; Eskew et al., 1981; Nayak et al., 1986), wheat (Bremer et al., 1995; Merbach et al., 1998), maize (Alexander and Zuberer, 1989), sorghum (Giller et al., 1984), foxtail millet (*Setaria italica*) (Okon et al., 1983) and various forage grasses (de Polli et al., 1977; Morris et al., 1985). In most of these studies ¹⁵N enrichment occurred predominantly in the rhizosphere and/or the roots, with very little translocation of ¹⁵N to the shoots/leaves, even after prolonged exposure to ¹⁵N₂-enriched atmospheres. This further suggests that much of the supposed plant-associated N₂ fixation is probably due to the uptake of N from mineralized free-living diazotrophs (Eskew et al., 1981; Lethbridge and Davidson, 1983; Okon et al., 1983; Nayak et al., 1986; Lee et al., 1994; Bremer et al., 1995). Certainly ¹⁵N₂ assays with shorter incubation times, or more frequent samplings over longer incubation periods, would help to distinguish between the more-or-less immediate transfer of fixed N on the one hand, and the longer term transfer of fixed N from dead and mineralized rhizosphere diazotrophs on the other. Better still, ¹⁵N₂ incorporation could be used in tandem with gas chromatography–mass spectrometry (GC–MS). This methodology has been successfully used to identify the immediate products of N₂ fixation in a number of symbioses including that between *Gunnera* and *Nostoc* (Silvester et al., 1996).

Few $^{15}\text{N}_2$ experiments have examined plants inoculated with specific diazotrophs. All such studies used *Azospirillum* spp., and showed little or no effect of inoculation on $^{15}\text{N}_2$ incorporation by rice, wheat, maize or foxtail millet (Nayak et al., 1986; Alexander and Zuberer, 1989; Okon et al., 1983; Merbach et al., 1998).

6. A strategy for obtaining evidence for symbiotic plant-associated N_2 fixation

6.1. Kallar grass and *Azoarcus*

The kallar grass–*Azoarcus* association is arguably the most extensively studied plant-associated “symbiosis”, and it provides a possible model of how to conduct studies into other putative symbioses. First, various field and greenhouse studies conducted by Malik and co-workers using ^{15}N isotope dilution and ^{15}N natural abundance have shown that kallar grass may fix up to 26% of its N content (Malik et al., 1997). Although a number of diazotrophs, including *Azospirillum* spp. can be isolated from the rhizosphere of kallar grass (Malik et al., 1997), the most abundant bacteria in surface-sterilized plant material appear to be *Azoarcus* spp. (up to $7.3 \times 10^7 \text{ g}^{-1}$ root dry weight; Reinhold-Hurek and Hurek, 1998a), suggesting that these organisms are at least partly responsible for the N_2 fixation. Further, microscope studies have shown immunogold-labelled colonies of *Azoarcus* within the vascular system and aerenchyma of kallar grass and rice (Hurek et al., 1994). More recent work has shown that these endophytic bacteria express *nif* genes and nitrogenase proteins *in planta* (Reinhold-Hurek and Hurek, 1998a,b, and see Section 4). Finally, laboratory experiments using the ^{15}N natural abundance technique with gnotobiotically grown kallar grass plantlets inoculated with either wild-type *Azoarcus* sp. BH72 or a *nif*- mutant have shown that the former incorporated significantly less ^{15}N than the latter (Hurek et al., 1998). This indicates that *Azoarcus* may fix N_2 *in planta* and transfer the fixed N to the host plant. The mechanism by which this transfer occurs has still to be determined, and it may simply be via the relatively inefficient process of the death and mineralization of the bacteria. Nevertheless, it is precisely this sort of data that are needed to bridge the gap between field/

greenhouse quantification studies on the one hand, and the isolation of specific diazotrophs on the other.

6.2. Sugar cane and *Acetobacter diazotrophicus*

Similar work on sugar cane/*A. diazotrophicus* is being undertaken in collaboration between the Department of Plant Pathology, University of Arizona and EMBRAPA-CNPAB, Rio de Janeiro. Sevilla et al. (1998), using sugar cane plantlets inoculated with wild type or *nif*- mutants of *A. diazotrophicus*, have shown that in the absence of fertilizer N, 30 days after inoculation, wild type bacteria significantly increased the height of plantlets above the uninoculated controls and those inoculated with the *nif*- bacteria. Interestingly, this effect was not observed when N was added to the plant growth medium. Further work by Sevilla and Kennedy (1999) has shown that at 60 days after inoculation there were significant differences in dry weight and total N content between the wild type and *nif*- inoculated plants, but again these were not significant when fertilizer N was added. Moreover, experiments using $^{15}\text{N}_2$ gas incorporation have shown that there can be significant uptake of fixed N by plants inoculated with *A. diazotrophicus*, and that N_2 fixation may contribute up to 0.5% of the plants total N over a 24 h period (Sevilla and Kennedy, 1999). However, as *nif*- mutants could also increase the growth and/or N content of sugar cane plants above that of the controls (Sevilla et al., 1998; Sevilla and Kennedy, 1999), it seems that a factor in addition to N_2 fixation is involved in the sugar cane–*A. diazotrophicus* association. This factor is most likely to be the production by *A. diazotrophicus* of plant growth hormones, such as indoleacetic acid (IAA) (Sevilla and Kennedy, 1999). Further studies are being undertaken to determine whether N_2 fixation or production of plant growth hormones by *A. diazotrophicus* makes the most important contribution to the nutrition of sugar cane (Kennedy, C., personal communication).

7. Conclusion

In the search for naturally occurring N_2 -fixing symbioses between non-legumes and associative/endophytic bacteria, James and Olivares (1998) advocated an integrated and progressive approach using

evidence for BNF from field experiments as the underlying basis of the research. However, although recent work with kallar grass/rice and with sugar cane has shown that this type of approach can indeed produce promising results, there is clearly much to be done in determining the organisms responsible for “symbiotic” BNF in grasses. It is essential that (1) sufficient numbers of diazotrophs should be present in the plants or in their rhizosphere to account for the observed BNF (2) that they can fix N_2 either *in planta* or in direct association with the plants, and (3) that they can benefit the plant via the transfer of fixed N products.

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